SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: CSA5101
COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Analytical Problem Solving: 40%

QUESTIONS: **Total Marks: 50**

Annexure A

Slot-A

Code of Conduct:

I Karanam Bhaskar Gnanesh (192324080) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

K.B.Gnanesh



Annexure A
Analytical Problem Solving

1. RSA Key Generation and Security Analysis
Given:
$p = 17$, $q = 19$
Compute n and $\phi(n)$
Choose $e = 5$ and calculate the private key d
Encrypt the message $M = 7$
Decrypt the ciphertext and verify correctness
Analyze the impact of choosing a small prime number on RSA security

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2. ElGamal Encryption Analysis
Given:
$p = 23$, $g = 5$, private key $x = 7$, random $k = 3$, message $M = 10$
Compute the public key
Perform encryption to generate ciphertext pair (C1, C2)
Decrypt the ciphertext to retrieve the original message
Analyze how randomness (k) improves security in ElGamal

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3. RSA Digital Signature Verification
Given:
$p = 11$, $q = 17$, $e = 7$, message $M = 8$
Compute n and $\phi(n)$
Find private key d
Generate the digital signature for the message
Verify the signature using the public key
Analyze how digital signatures ensure non-repudiation

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4. Diffie-Hellman with Attack Resistance Evaluation
Given:
$p = 29$, $g = 2$
User A private key = 5
User B private key = 12
Compute public keys for both users
Compute the shared secret key
Analyze what happens if an attacker intercepts and replaces public keys
Suggest a cryptographic solution to ensure authenticity

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5. Hybrid Encryption System Analysis
Given:
A system uses RSA for key exchange and AES for data encryption
RSA encryption time = 5 ms per key exchange
AES encryption time = 0.5 ms per data block
Total data blocks = 500
Calculate total time taken for encryption
Compare with using only RSA for encrypting all data

Analyze why hybrid encryption is more efficient
Justify its use in modern secure communication systems

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20% 20	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20% 20	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25% 20	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process

Accuracy of Calculation	20% 20	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15% 10	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

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1. RSA Key Generation and Security analysis

Sol

Given

$p = 17, \quad q = 19$

Public exponent $e = 5$

message : $M = 7$

$n = pq = 17 \times 19 = 323$

$n = 323$

$\phi(n) = (p-1)(q-1)$

$\phi(n) = 16 \times 18 = 288$

$ed = 1 \bmod \phi(n)$

$5d = 1 \bmod (288)$

$288 = 5(57) + 3$

$5 = 3(1) + 2$ $\qquad 3 = 2(1) + 1$

$1 = 2(288) - 155(5)$

$d = -155 = 173 \bmod (288)$

$C = M^e \bmod n$

$C = 7^5 \bmod 323$

$C = (140 \times 7) \bmod 323$

$\boxed{C = 10}$

$M = C^d \bmod n$

$M = 10^{173} \bmod 323$

$\boxed{M = 7}$

$C_1 = g^k \bmod P$

$C_1 = 5^3 \bmod 23 \quad = 10$

$\boxed{C_1 = 10}$

$C_2 = M \times y^k \bmod P$

$y^k = 17^3 \bmod 23$

$17^2 = 289 = 13 \bmod 23 \quad = 14 \bmod 23$

$C_2 = (10 \times 14) \bmod 23$

$\boxed{C_2 = 2}$

final ciphertext

$(C_1, C_2) = (10, 2)$

2. **El Gamal Encryption Analysis**

Given

Prime $P = 23$

Generator $g = 5$

Private key $x = 7$

Random key $k = 3$

message $M = 10$

$$y = g^x \bmod p$$

$$y = 5^7 \bmod 23$$

$$5^2 = 25 = 2 \pmod{23}$$

$$5^4 = 2^2 = 4 \pmod{23}$$

$$5^7 = 5^4 \times 5^2 \times 5$$

$$y = (4 \times 2 \times 5) \bmod 23 = 17$$

$$\boxed{y = 17}$$

Public key $(P, g, y) = (23, 5, 17)$

Private key $x = 7$

$C_1 = g^k \bmod p$

$C_1 = 5^3 \bmod 23 = 10$

$\boxed{C_1 = 10}$

$C_2 = M \times y^k \bmod p$

$y^k = 17^3 \bmod 23$

$17^2 = 289 = 13 \pmod{23}$

$17^3 = 13 \times 17 = 221 = 14 \pmod{23}$

$C_2 = (10 \times 14) \bmod 23$

$\boxed{C_2 = 2}$

final ciphertext :

$(C_1, C_2) = (10, 2)$

$S = C_1^x \bmod p$

$S = 10^7 \bmod 23$

$\boxed{S = 14}$

$M = C_2 S^{-1} \bmod p$

$M = (2 \times 5) \bmod 23$

$\boxed{M = 10}$

RSA Digital Signature verification

$P = 11,\ q = 17$

Public exponent $e = 7$

Message $M = 8$

n & $\phi(n)$

$$n = pq = 11 \times 17 = 187$$

$$\boxed{n = 187}$$

$$\phi(n) = (p-1)(q-1)$$

$$= 10 \times 16$$

$$\boxed{\phi(n) = 160}$$

$$ed = 1 \bmod (\phi(n))$$

$$7d = 1 \bmod (160)$$

$$161 \bmod 160 = 1$$

$$\boxed{d = 23}$$

$S = M^d \bmod n$

$S = 8^{23} \bmod 187$

$8^2 = 64$

$8^4 = 64^2 \bmod 187 = 169$

$8^8 = 169^2 \bmod 187 = 137$

$8^{16} = 137^2 \bmod 187 = 69$

$S = (69 \times 169 \times 64 \times 8) \bmod 187$

$\boxed{S = 184}$

$M' = S^e \bmod n$

$M' = 184^7 \bmod 187$

$\boxed{M' = 8}$

Diffie - Hellman with Attack Resistance Evaluation

Prime $P = 29$

generator $g = 2$

user A's Private key $a = 5$

user B's Private key $b = 12$

$$A = g^a \bmod P$$

$$A = 2^5 \bmod 29$$

$$A = 32 \bmod 29$$

$$\boxed{A = 3}$$

$$B = g^b \bmod P$$

$$B = 2^{12} \bmod 29$$

$$2^{12} = 4096$$

$$B = 4096 \bmod 29$$

$$\boxed{B = 7}$$

$K_A = B^a \bmod p$

$K_A = 7^5 \bmod 29$

$7^2 \bmod 29 = 20$

$7^4 \bmod 29 = 23$

$K_A = B^a \bmod p$

$K_A = 7^5 \bmod 29$

$7^2 \bmod 29 = 20$

$7^4 \bmod 29 = 23$

$K_A = (23 \times 7) \bmod 29 = 16$ $\boxed{K_A = 16}$

$K_B = A^b \bmod P$

$= 3^{12} \bmod 29$

$\boxed{K_B = 16}$

$\boxed{K_A = K_B = 16}$

5. Hybrid Encryption System analysis

RSA = 5ms

AES = 0.5ms

Total data blocks = 500

$T_{AES} = 500 \times 0.5$

$T_{AES} = 250$ ms

$T_{hybrid} = T_{RSA} + T_{AES}$

$T_{hybrid} = 5 + 250$

$T_{hybrid} = 255$ ms

$T_{RSA\ only} = 500 \times 5$

$= 2500$ ms

The hybrid system saves

$2500 - 255 = 2245$ ms

$\frac{2500 - 255}{2500} \times 100 = 89.8\%$